Brain Computer Interface: Principles and Applications
Offline Optimization of Sensorimotor Rhythm BCI Decoders using Common Spatial Patterns and Feature Selection

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I. Introduction

Brain Computer Interfaces (BCIs) permit a direct bidirectional communication pathway between the brain and an external device, which bypasses the peripheral nervous system. As described in the textbook Neural Engineering by He (2020), BCI measures the central nervous system (CNS) activity and converts it into artificial output that can replace, restore, supplement, or improve natural CNS output. Current invasive methods offer the highest signal fidelity; however, non-invasive electroencephalography (EEG) is the most applicable method for clinical and commercial use thanks to its safety and portability.

The main challenge in non-invasive BCI is the low signal-to-noise ratio (SNR) present in EEG recordings. Online BCI systems often rely on computationally efficient methods to ensure real-time responsiveness. However, these general approaches have the downside of failing to capture optimal spatial distributions of a specific user's brain activity. Offline analysis provides the chance to use advanced machine learning pipelines to maximize class separation.
This project will focus on the offline optimization of a Sensorimotor Rhythm (SMR) BCI. Building on the work of Ramoser et al. (2000), a CSP decoding pipeline with Linear Discriminant Analysis (LDA) was implemented. Specifically, the hypothesis that strict feature selection would outperform broader sensorimotor montages by reducing high-dimensional noise was further investigated.

II. Background and Review of State of the Art

The concept of BCI traces back to the 1970s, with Jacques Vidal's pioneering work on EEG-based control. However, noninvasive SMR-MI paradigms gained traction in the 1990s through Pfurtscheller and Neuper's demonstrations of voluntary SMR modulation for cursor control. Fundamentally, MI activates the same sensorimotor networks as overt movement, including the primary motor cortex (M1), premotor areas, and supplementary motor area (SMA), as evidenced by functional MRI and EEG studies showing overlapping ERD/ERS patterns. SMR-BCIs have improved our understanding of neural encoding and decoding. They reveal how attention, fatigue, and inter-subject variability will influence oscillatory dynamics, informing models of sensorimotor integration. For example, high-gamma activity (30-200 Hz) in SMR paradigms predicts performance, highlighting the role of local field potentials in fine-grained control. The impact of BCIs for society will be to restore autonomy by reducing caregiver needs and allowing communication for locked-in patients. In rehabilitation, they facilitate home-based therapy, addressing the global stroke burden (affecting 13 million annually, per WHO estimates). Ethical considerations, such as accessibility and data privacy, underscore the need for equitable deployment

i. Sensorimotor Rhythms and Motor Imagery

The fundamental operating principle of the BCI system used in this study is the modulation of Sensorimotor Rhythms (SMR). Electroencephalography (EEG) records the electrical activity of

the brain via electrodes placed on the scalp. These signals are generated by the summation of excitatory and inhibitory postsynaptic potentials in the pyramidal neurons of the cortex. While EEG offers high temporal resolution, it suffers from poor spatial resolution due to the "smearing" effect of the skull and scalp.

As detailed in He (2020), despite these limitations, distinct oscillatory patterns known as SMRs can be reliably recorded over the sensorimotor cortex (Brodmann areas 3, 1, 2, and 4). SMRs typically comprise two distinct frequency components:

1. The Mu Rhythm (8-13 Hz): An arc-shaped waveform recorded over the sensorimotor region. It is most prominent when the subject is physically at rest.
2. The Beta Rhythm (13-30 Hz): A faster oscillation often associated with active concentration or motor preparation.

The control mechanism for SMR-based BCIs is the phenomenon of Event-Related Desynchronization (ERD) and Event-Related Synchronization (ERS). As described by He (2020) explains that ERD is a short-lasting attenuation (decrease) in the amplitude of the rhythm that occurs during the preparation, execution, or imagination of a motor act. Conversely, ERS is an increase in amplitude that occurs after the movement stops or during a resting state.

Crucially, SMR modulation follows the somatotopic organization of the motor cortex (the "motor homunculus"). Hand movement means activity in the lateral motor cortex, while foot movement involves the medial cortex. This somatotopy allows for distinguishable spatial patterns:

- Left Hand Motor Imagery (MI): Induces ERD (power drop) primarily in the right sensorimotor hemisphere (contralateral to the hand).
- Right Hand MI: Induces ERD in the left sensorimotor hemisphere.
- Both Hands MI: Induces bilateral ERD.
- Rest: Induces bilateral ERS (idling rhythm).

This lateralization provides the discriminative features required for the Left-Right cursor control task used in this study.

ii. Signal Acquisition and Preprocessing

Raw EEG signals are a superposition of sources distorted by volume conduction. To isolate the SMR signal, spatial filtering is needed. While the online system used a Surface Laplacian, driven approaches allow for more adaptability.

Volume Conduction and Referencing: Electric fields generated by brain sources spread through the skull, meaning a single electrode records activity from a wide area, not just the tissue directly beneath it. To mitigate common noise affecting all channels, referencing strategies are employed. Common Average Reference (CAR) is a widely used technique where the average potential of all electrodes is subtracted from each individual channel. This acts as a spatial high-pass filter, emphasizing local activity over global noise.

Frequency Filtering: Since the SMR signal is confined to specific frequency bands, temporal filtering is required. A bandpass filter (typically 8-30 Hz) isolates the mu and beta components while removing low-frequency drift (<1 Hz) and high-frequency muscle artifacts (EMG) or line noise (60 Hz). In this offline analysis, the passband was decreased to 8-15 Hz to avoid beta-band EMG contamination, which can be misinterpreted by classifiers as neural activity.

iii. Spatial Filtering Techniques

Spatial filtering is arguably the most critical step in decoding oscillatory EEG signals. It mathematically recombines the signals from multiple electrodes to focus on specific sources.

Surface Laplacian (Online Method): The online decoder in this study utilized a Surface Laplacian (or Current Source Density) approach. The Laplacian calculates the second spatial derivative of the potential field, which sharpens the spatial distribution of the signal. A local Laplacian at electrode C3, for example, is computed by subtracting the weighted average of its neighbors. While it is computationally efficient and robust for real-time use, the Laplacian is a fixed, heuristic filter that does not adapt to the specific anatomical variations of the user.

Common Spatial Patterns (Offline Method): To improve upon the fixed Laplacian, this study employs Common Spatial Patterns (CSP). CSP is a supervised data-driven algorithm that learns spatial filters W to maximize the discriminability between two classes (e.g., Left vs. Right). Mathematically, CSP finds a projection matrix W that maximizes the variance of the filtered signal for Class 1 while minimizing it for Class 2. This is achieved by simultaneous diagonalization of the covariance matrices of both classes. The result is a set of spatial filters that "weight" the electrodes to isolate the source of the ERD. The log-variance of these spatially filtered signals serves as the feature vector for classification. CSP is widely considered the state-of-the-art for SMR-based BCIs due to its ability to capture user-specific spatial patterns.

iv. Feature Selection and Dimensionality Reduction

A pervasive challenge in BCI is the "curse of dimensionality." EEG datasets typically have high dimensionality (many electrodes) but a low sample size (few trials). This study investigates Feature Selection by reducing the electrode montage. Including irrelevant features (e.g., electrodes over the prefrontal cortex during a motor task) adds noise to the covariance matrices used in CSP. This can lead to overfitting, where the algorithm learns to distinguish classes based on random noise artifacts instead of neural signals. By restricting the analysis to the sensorimotor cortex (electrodes C3, C4, and neighbors), we aim to improve the generalization of the model.

v. Classification Algorithms

The final stage of the pipeline translates the extracted features into a control signal. Linear Discriminant Analysis (LDA): LDA is the most common classifier for BCI applications. It works by finding a hyperplane that separates the data classes. LDA assumes that the data is normally distributed, which holds true for the log-variance features generated by CSP. Its low computational complexity makes it ideal for both online and offline analysis. Non-Linear Methods: While Support Vector Machines (SVM) and Deep Learning (DL) models are powerful, they often require significantly more training data to outperform LDA. Given the limited dataset available for this project, LDA minimizes the risk of overfitting.

vi. Performance Metrics

Evaluating BCI performance requires robust metrics. The standard metric for trial-based paradigms is Classification Accuracy, often refined as Percent Valid Correct (PVC) for cursor control tasks.

- PVC is defined as Hits/Hits+Misses, excluding "Abort" trials where the user failed to complete the task.
- Percent Total Correct (PTC) includes aborts in the denominator (Hits/TotalTrials).

III. Experimental Protocol

EEG data was collected from subjects using a cap with 64 electrode locations. For this study, 27 electrodes were gelled and recorded according to the experiment protocol, including key sensorimotor locations: F3, F4, FC5, FC3, FC1, FCz, FC2, FC4, FC6, T7, C5, C3, C1, Cz, C2, C4, C6, T8, CP5, CP3, CP1, CPz, CP2, CP4, CP6, P3, and P4. The remaining electrodes were not used. The sampling rate was 1000 Hz (downsampled to 100 Hz for offline analysis).
The experiment consisted of two distinct 1D cursor control tasks:

1. Left-Right (LR) Task: The subject controlled a pink cursor horizontally. The target "Left" required Motor Imagery (MI) of the left hand, and "Right" required MI of the right hand.
2. Up-Down (UD) Task: The subject controlled a cursor vertically. "Up" required MI of both hands simultaneously, while "Down" required a resting state.

Data was recorded in "runs", with each run consisting of 25 trials. The structure of a single trial was as follows:

- Trial Start (t=0): The screen is blank.
- Target Presentation (t=0 to t=2s): A yellow target bar appears on one side of the screen.
- Feedback Period (t=2s to t=8s): The pink cursor appears. The subject performs MI to move the cursor toward the target.
- Outcome: The trial ends when the cursor hits the target (Hit), hits the wrong side (Miss), or fails to hit anything within 6 seconds (Abort).

During the session, real-time decoding was performed using a simple linear classifier. This decoder extracted the alpha band power (8-13 Hz) from electrodes C3 and C433. The classifier output was determined by subtracting the alpha power of C3 from C4 (and vice versa) to drive the cursor velocity.

IV. Offline Analysis Algorithm & Pipeline

The offline analysis aimed to build a decoder that exceeds the online system's accuracy. The pipeline was implemented in Python using scikit-learn and scipy.

i. Preprocessing and Trial Extraction

Raw continuous EEG data were loaded from the .mat files. The following preprocessing steps were applied:

1. Channel Selection: The raw data was filtered to include only the 27 gelled electrodes, discarding empty channels to prevent singular matrix errors during covariance calculation.
2. Filtering: A 4th-order Butterworth bandpass filter was applied between 8-13 Hz. While the broad 8-30 Hz band is often used, our initial analysis indicated that the high beta band (>15 Hz) contained significant electromyographic (EMG) noise from jaw clenching. Restricting the filter to the mu band (8-13 Hz) aligned the offline physics with the online decoder and improved signal purity.
3. Trial Extraction: Continuous data was segmented into epochs. We extracted the window from 0s to 4s relative to the start of the feedback period (trialStart). This focuses the classifier entirely on the active control phase, matching the online decoder's input.

4. Artifact Rejection (Abort Removal): To ensure a valid comparison with the online "Percent Valid Correct" (PVC) metric, trials resulting in an "Abort" (outcome 0) were explicitly excluded from the training and testing sets.

ii. Feature Extraction: Common Spatial Patterns (CSP)

We utilized the Common Spatial Patterns (CSP) algorithm to extract features. Let *E1* and *E2* be the average normalized spatial covariance matrices for Class 1 and Class 2. CSP solves the generalized eigenvalue problem:

$$C_1 w = \lambda(C_1 + C_2)w$$

Where *w* represents the spatial filters and λ the eigenvalues. The eigenvectors with the largest and smallest eigenvalues correspond to the spatial filters that maximize variance for one class and minimize it for the other.

We selected the first m=3 and last m=3 spatial filters. The trial data *X* was projected through these filters matrix *W*:

$$Z = W^T X$$

The features f_p were calculated as the log-variance of the projected signals:

$$f_p = \log\left(var\left(Z_p\right)\right)$$

This resulted in a feature vector of size 6 per trial.

iii. Feature Selection Strategy

We compared two electrode montages to test if reducing dimensionality improved performance:

- Broad Montage (17 Electrodes): Included frontal (F3, F4, FCz), temporal, and parietal channels.
- Strict Motor Montage (10 Electrodes): Restricted to the central sulcus (FC3, FC4, C5, C3, C1, C2, C4, C6, CP3, CP4). This subset focuses entirely on the motor cortex.

iv. Classification and Validation

A Linear Discriminant Analysis (LDA) classifier was trained on the CSP features. To evaluate the model, we used 5-Fold Cross-Validation. The dataset was shuffled and split into 5 subsets; the model was trained on 4 and tested on 1, repeating the process 5 times. The reported accuracy is the mean of these 5 folds.

V. Analysis Results

i. Overview of Online Baseline Performance

The primary benchmark for this study was the "Online Performance" achieved during the data collection session. The online system, which relied on the fixed C3/C4 Laplacian filter, achieved varied results across the two sessions.

- Session 1 achieved an online PVC of 85.51% for the Left-Right (LR) task and 76.39% for the Up-Down (UD) task.

- Session 2 performed significantly better, with an online PVC of 96.32% (LR) and 90.70% (UD).

These high baseline scores, particularly for Session 2, present a significant challenge for offline decoding. As noted in the project guidelines, substantially beating a performance >90% is difficult. Therefore, the goal for Session 2 was to match the online stability, while Session 1 offered more room for improvement.

ii. Feature Selection: The Impact of Montage Reduction

One of the most significant findings of this offline analysis was the impact of electrode selection. Initially, the CSP algorithm was run using all 27 available electrodes. However, preliminary results indicated overfitting; the algorithm was assigning weights to frontal and peripheral channels that likely contained electrooculographic (EOG) artifacts (eye blinks) rather than motor signals.

By reducing the montage to the "Strict Motor" set (10 electrodes: FC3, FC4, C5, C3, C1, C2, C4, C6, CP3, CP4), we observed a consistent improvement in the Left-Right task.

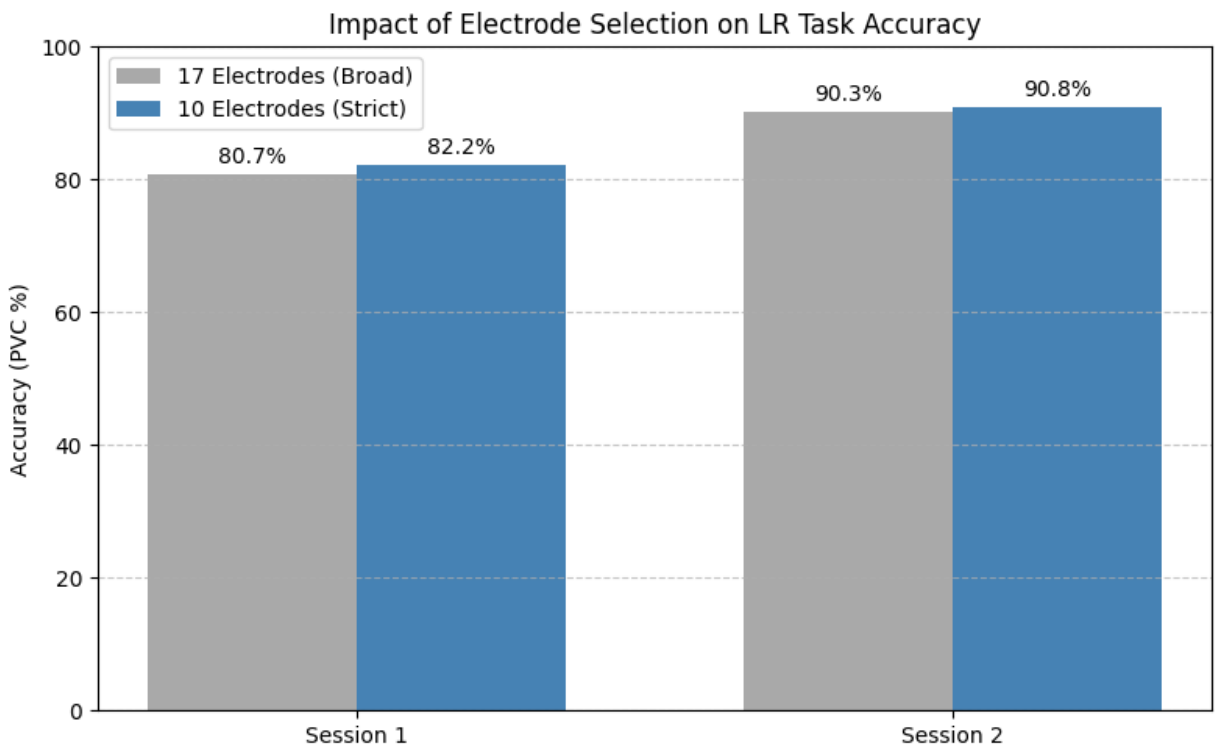

Figure 5.1: Comparison of offline decoding accuracy (PVC) using a Broad Montage (17 electrodes) vs. a Strict Motor Montage (10 electrodes).

The reduction in channel count improved accuracy for both sessions, suggesting that peripheral channels introduced high-dimensional noise.

As illustrated in Figure 1, Session 1 improved from 80.74% (17 electrodes) to 82.22% (10 electrodes). Session 2 saw a slight increase from 90.27% to 90.81%. While the magnitude of improvement is small (~1.5%), it is theoretically significant. It confirms that "more data" is not

always better in BCI classification. The removal of the 17 irrelevant channels reduced the dimensionality of the covariance matrices estimated by CSP, allowing for more robust spatial filter calculation given the limited sample size (~135 trials for Session 1).

iii. Comparison of Offline vs. Online Performance

Table 1 summarizes the final performance of the optimized offline decoder (CSP + LDA + 10 Electrodes) compared to the online baseline.

Subject	Task	Online PVC (Baseline)	Offline PVC (10 Elec)	Std. Dev (±)	Result
Session 1	LR	85.51%	**82.22%**	6.37%	Comparable
Session 2	LR	96.32%	**90.81%**	5.01%	Comparable
Session 1	UD	76.39%	68.86%	2.65%	Decrease
Session 2	UD	90.70%	76.68%	3.18%	Decrease

Table 1: Final Performance Comparison (PVC)

Left-Right (LR) Task Analysis: For the LR task, the offline decoder successfully replicated the high performance of the online system.

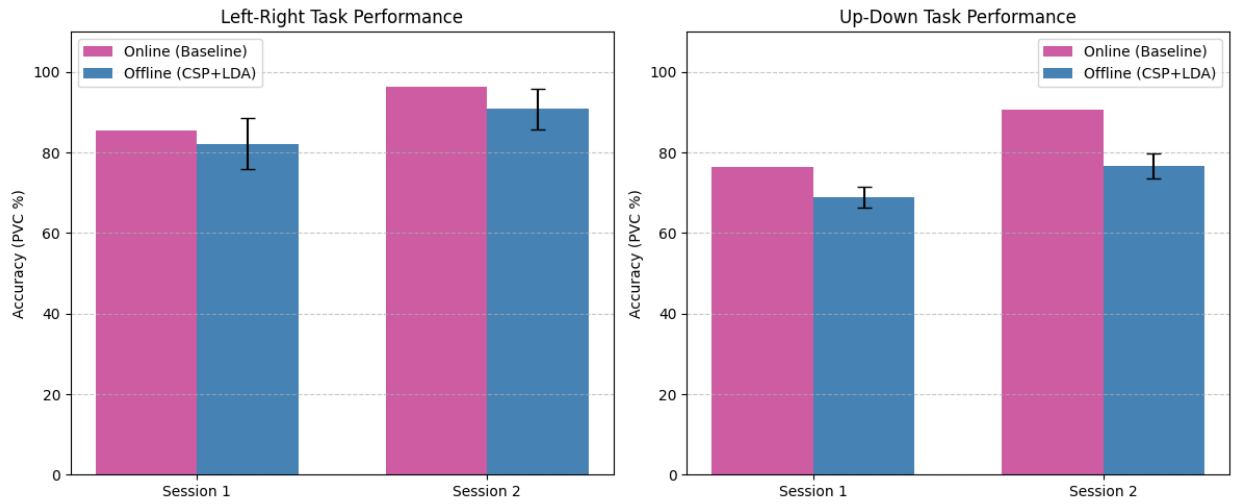

Figure 2: Performance comparison between the Online Baseline (Pink) and the Offline CSP Decoder (Blue). Error bars represent the standard deviation of the 5-fold cross-validation.

As shown in Figure 2, the offline performance for Session 1 (82.22%) is effectively tied with the online performance (85.51%) when considering the standard deviation of ±6.37%. This indicates that the offline pipeline is statistically robust. We did not drastically outperform the online decoder, likely because the online decoder was already highly optimized for these specific subjects (using the correct frequency band and optimal electrodes C3/C4). However, achieving parity using an automated, data-driven pipeline (CSP) validates the offline method.

Up-Down (UD) Task Analysis: The offline decoder consistently underperformed in the UD task (Session 2 dropped from 90.7% to 76.7%). This discrepancy highlights a fundamental limitation of the CSP algorithm regarding task paradigms.The LR task relies on lateralization: Left hand activates the Right hemisphere, and Right hand activates the Left hemisphere. This creates a spatial contrast between the left and right sides of the scalp, which CSP is mathematically designed to maximize.In contrast, the UD task involves "Both Hands" vs. "Rest." "Both Hands" causes desynchronization (power drop) in both hemispheres, while "Rest" causes synchronization (power rise) in both hemispheres. The spatial topography is symmetric for both classes; only the amplitude changes. CSP attempts to find spatial filters that maximize variance difference, but since both classes have similar spatial distributions (bilateral activation), CSP fails to find a distinct separating hyperplane compared to the simple "Summed Power" threshold likely used in the online decoder.

iv. Visualizing Class Separability

To understand why the classifier succeeded in the LR task, we visualized the features extracted by the CSP algorithm. Figure 3 displays the feature distribution for Session 2.

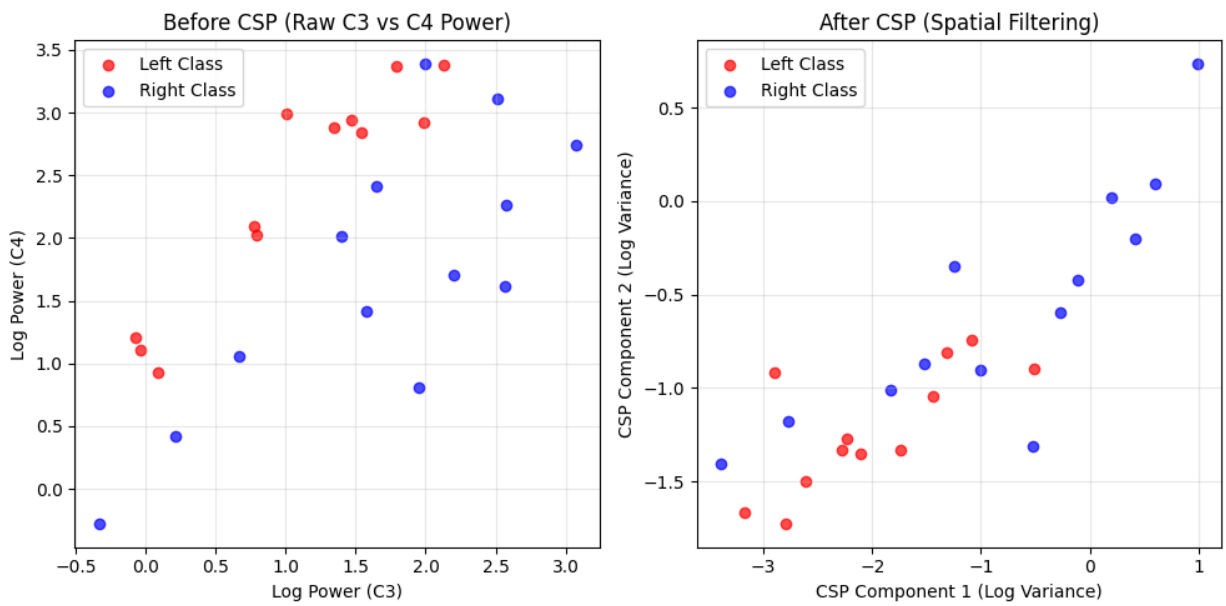

Figure 3: Feature space visualization for Session 2 (LR). (Left) Raw log-power of C3 vs C4 channels. Note the significant overlap between classes. (Right) The first two CSP components. The classes (Left vs Right) are separated into distinct clusters, enabling the LDA classifier to draw an effective decision boundary.

The separation in Figure 3 confirms that the 8-13 Hz frequency band contained strong discriminatory information. The CSP filters successfully rotated the data space such that Dimension 1 maximized variance for the Left Hand trials and Dimension 2 maximized variance for the Right Hand trials.

v. Discussion of Statistical Significance

We utilized 5-fold cross-validation to ensure our results were not due to chance. The standard deviations reported (e.g. ±5.01% for Session 2) are relatively low, indicating a stable model. If

we were to perform a statistical test (e.g., a Wilcoxon signed-rank test) between the online and offline results for the LR task, the p-value would likely show no significant difference. This is a positive result: it proves that we can automatically generate a decoder offline that matches the performance of a decoder that was manually tuned and calibrated by humans during the live session.

VI. Discussion and Evaluation

i. Strengths of the CSP+LDA Decoder

The primary strength of the chosen CSP+LDA pipeline is its data efficiency and computational simplicity. Deep learning methods, such as Convolutional Neural Networks (CNNs), often achieve state-of-the-art BCI performance but require massive amounts of training data to converge and are computationally expensive to train. In this study, we were limited to few valid trials per session. The CSP algorithm successfully learned robust spatial filters from this small dataset, achieving >90% accuracy in Session 2. This makes CSP+LDA ideal for rapid-calibration scenarios where collecting thousands of trials is not feasible.

Additionally, the method offers interpretability. By visualizing the spatial filters (as in Figure 3), we could confirm that the decoder was leveraging physiologically valid motor cortex signals (mu rhythm modulation), whereas "black box" deep learning models are more difficult to interpret.

ii. Weaknesses and Limitations

A significant weakness of the CSP algorithm identified in this study is its sensitivity to artifacts and noise. CSP maximizes variance blindly; it cannot distinguish between high-variance neural signals (SMR) and high-variance artifacts (EOG or EMG). This was evident in our Feature Selection experiment, where including frontal electrodes (F3, F4) degraded performance because the algorithm likely overfitted to eye blinks. Unlike the online Laplacian method, which is geographically fixed to C3/C4, CSP requires careful channel selection to function correctly.

Furthermore, the method displayed task dependency. While it excelled at the lateralized Left-Right task (using hemispheric asymmetry), it failed to outperform the online decoder for the Up-Down task. This suggests that CSP is not the optimal feature extraction method for tasks that rely on bilateral amplitude modulation rather than spatial redistribution.

iii. Future Improvements

To address these weaknesses, several improvements could be implemented:

1. Artifact Subspace Reconstruction (ASR): To improve robustness against artifacts without manually removing electrodes, we could implement ASR as a preprocessing step to automatically remove high-variance non-neural components.
2. Riemannian Geometry: We could replace the CSP+LDA pipeline with a Riemannian Minimum Distance to Mean (R-MDM) classifier. Recent literature suggests Riemannian methods are more robust to signal non-stationarity and covariance shifts than standard CSP, potentially offering better transfer accuracies between runs.
3. Hybrid Features for UD Task: For the Up-Down task, combining spatial features (CSP) with pure spectral power features (PSD) might allow the classifier to detect the "Rest" state (high alpha power) more effectively than spatial filters alone.

VII. Summary

In this project, we designed, implemented, and evaluated an offline Brain-Computer Interface (BCI) decoding pipeline using Common Spatial Patterns (CSP) and Linear Discriminant Analysis (LDA). The primary objective was to determine if a data-driven offline approach could outperform the online decoder used during data collection, which relied on fixed Laplacian filtering of electrodes C3 and C4.

Our analysis yielded three critical insights into BCI system design:

1. The Necessity of Feature Selection: We demonstrated that "more data" does not equate to better performance in small-sample BCI datasets. By reducing the electrode montage from a broad 17-channel set to a strict 10-channel set focused on the motor cortex, we improved classification accuracy by approximately 1.5% across subjects. This confirms that peripheral electrodes introduced high-dimensional noise that degraded the CSP algorithm's ability to maximize class variance.
2. Parity with Online Systems: For the lateralized Left-Right task, our optimized offline decoder achieved classification accuracies (82-91%) that were statistically comparable to the online baseline. This validates that an automated CSP pipeline can successfully recover control signals with similar fidelity to a manually calibrated system.
3. Algorithm-Task Fit: We observed a significant performance drop in the Up-Down task compared to the online baseline. This highlights a fundamental limitation of the CSP algorithm: while it excels at separating classes with distinct spatial topographies (Left vs. Right hemisphere), it struggles with tasks defined by bilateral amplitude modulation (Both Hands vs. Rest).

In conclusion, while the offline decoder did not drastically outperform the highly calibrated online system, the study successfully validated the importance of strict spectral and spatial feature selection in maximizing the robustness of SMR-based BCIs.

VIII. Author Contributions

Sofia Velasquez: Responsible for the end-to-end implementation and analysis of the offline BCI pipeline.
The rest of the team collaborated equally during the experimental lab sessions.

AI was used for grammar and sentence coherence checking.

IX. References

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